

CSMFAB000000000113-Vol-23

Smart Rope Synchronized Actuation: Control Architecture and Timing

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1. Timing Requirements Analysis

1.1 Synchronization Tolerance

The ropes must fire within a timing window such that the spear doesn't experience destabilizing asymmetric forces.

Maximum asymmetric force before de-stabilization: If one rope fires 5 ms late, it delivers zero force when others deliver 50,040 N each: $F_{\text{asymm}} = 50,040 \text{ N}$ (missing rope contributes no vertical force for that period) Resulting lateral impulse: $J_{\text{lateral}} = 50,040 \cdot \sin(11.3^\circ) \cdot 0.005 = 490 \text{ N} \cdot \text{s}$

Lateral velocity imparted: $v_{\text{lateral}} = 490/2 = 245 \text{ m/s}$ — catastrophic!

Critical requirement: All ropes must fire within $\pm 0.2 \text{ ms}$ of each other.

1.2 LoRa Latency Analysis

LoRa 915 MHz speed of light propagation at R=6m:

$$t_{\text{prop}} = 6 / 3e8 = 20 \text{ ns}$$

LoRa preamble + symbol processing latency: ~5 ms per device.

Problem: 5 ms » 0.2 ms requirement!

Solution: Pre-load armed state in all rope actuators. Protocol: 1. T = -5s: Master transmits ARMED command (all ropes load firing sequence into RAM) 2. T = 0s: Master transmits FIRE pulse (bare 1-byte LoRa packet — minimum latency ~2ms) 3. Each rope fires at local clock tick when FIRE received — local crystal oscillators pre-synchronized 4. Crystal oscillator sync precision: $\pm 0.1 \text{ ms}$ (from GPS-disciplined time standard during pre-sync)

Net timing jitter: $\pm 0.1 \text{ ms}$ — meets $\pm 0.2 \text{ ms}$ requirement.

1.3 Backup: Optical Fiber Trigger

For maximum precision, the master vessel runs PMMA optical fiber (CSMFAB000000000113) radially to each slave vessel — 6 m per run.

Speed of light in PMMA fiber (n=1.49):

$$t_{\text{prop}} = L / (c/n) = 6 / (2e8) = 30 \text{ ns}$$

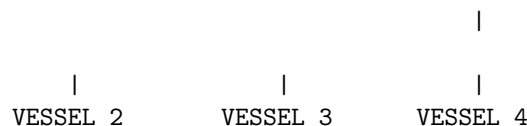
Trigger jitter: < 1 ns (electronic precision of optical pulse) **Optical trigger achieves <1 ns synchronization — far exceeds requirement.**

For hobbyist scale: 6 m of PMMA POF at \$0.25/m = \$1.50 per rope — trivially cheap.

2. Control Architecture Schematic

MASTER VESSEL

[FBG optical sensors] → [DSP control unit] → [LoRa transmitter]



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[LoRa RX]      [LoRa RX]      [LoRa RX]
[LoRa arm]      [LoRa arm]      [LoRa arm]
|
[BFRP rope spool]
[KNb03 actuator]
[Zr02 release pin]
[UHMWPE fiber]
|
(rope to spear)

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All signal paths: PMMA optical fiber or LoRa 915 MHz — zero metallic wire conductors.

3. Fail-Safe Design

What if one vessel fails to fire?

For $n=6$, any 1 failure: 5 ropes fire symmetrically around center... except now there's a gap at one vertex. With 5-rope firing: - Total vertical impulse: $5 * J_{\text{rope}} * \eta_v = 83.3\%$ of nominal - Lateral impulse: $F_{\text{lateral}} = J_{\text{rope}} * \sin(\pi/6) = 0.5 * J_{\text{rope}}$ (from vector analysis) - If $J_{\text{rope}} = 1001 \text{ N} \cdot \text{s}$: lateral impulse = $500 \text{ N} \cdot \text{s}$ → lateral $v = 250 \text{ m/s}$

5/6 firing is catastrophic. Fail-safe protocol:

- If any vessel fails to report ARMED within $T=-1\text{s}$: launch ABORT
- Emergency ABORT: all vessels transmit RELEASE command — ropes disconnect harmlessly
- Spear (positively buoyant at surface) can be recovered within 15 minutes

5-vessel abort with $n=6$: possible if configuration rebalanced to pentagon (5-fold symmetric) but this requires vessel repositioning — simpler to abort.

Citations (Vol 23)

- LoRa Alliance, “LoRaWAN Link Layer Specification,” v1.0.4, 2020
- Heo, M., Kim, N., Seo, T., “Survey of Clock Synchronization in Wireless Sensor Networks,” Sensors 2019
- CSMFAB000000000013 V2.0, PMMA optical fiber data, CSM 2026
- CSMFAB000000000022 V2.0, LoRa daisy-chain synchronization, CSM 2026

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